

TripNest

1. Title

TripNest: Travel Planning & Trip Management Platform

2. Objective

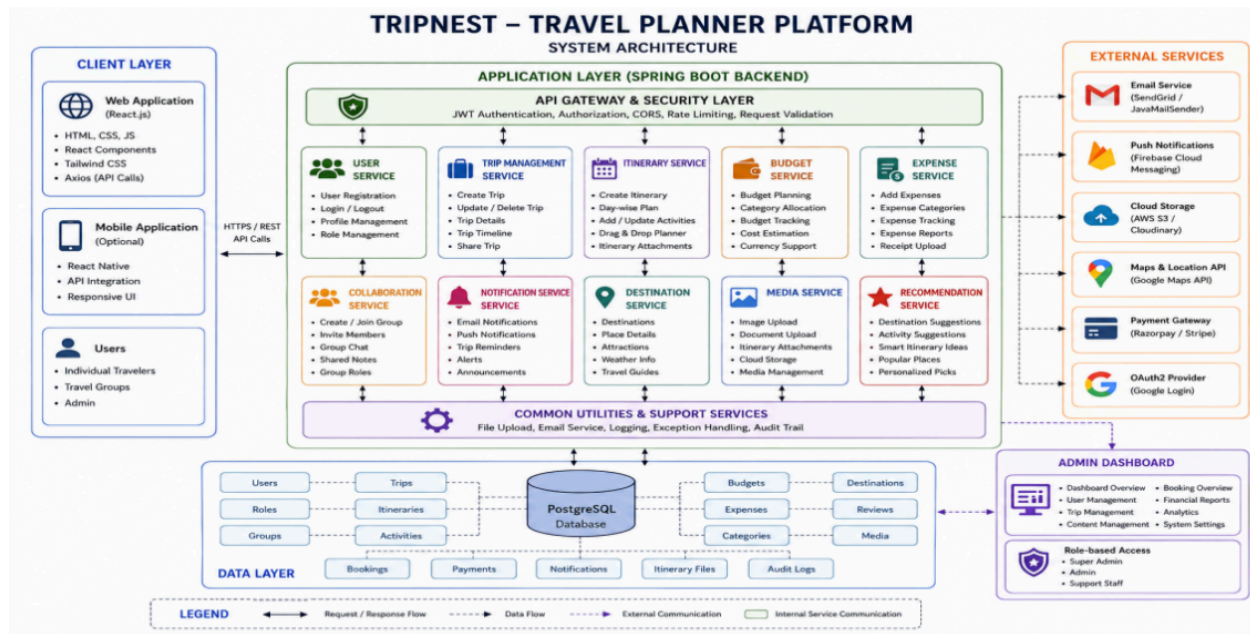
Build a full-stack web application (React.js frontend + Spring Boot backend) that enables travelers to plan trips, create day-wise itineraries, manage travel budgets, collaborate with friends and family, track expenses, and organize travel documents through a centralized travel dashboard.

The platform provides personalized travel planning, trip collaboration, budget management, destination discovery, and travel analytics features.

Outcomes

- Developed and deployed a full-stack React.js + Spring Boot application.
 - Implemented secure user authentication using JWT and OAuth2.
 - Built trip creation and itinerary management functionality.
 - Implemented budget and expense tracking systems.
 - Added collaborative trip planning and group management.
 - Created dashboards for trip analytics and expense reports.
 - Integrated maps and location services.
 - Deployed frontend and backend using PostgreSQL in production.
 - Added notifications for trip reminders and travel updates.
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3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login
- Password Reset
- Profile Management

Roles

- Traveler
- Group Admin
- Administrator

2. User Profile Management

- Profile creation
- Travel preferences
- Favorite destinations

- Travel history
- Account settings
- Profile customization

3. Trip Management System

- Create trips
- Edit trips
- Delete trips
- View trip details
- Trip sharing
- Trip timeline

Trip Information

- Destination
- Start Date
- End Date
- Travelers
- Budget
- Status

4. Itinerary Planning System

- Day-wise itinerary creation
- Activity scheduling
- Place management
- Travel timeline
- Activity reminders

Activity Types

- Sightseeing
- Transportation
- Accommodation
- Dining
- Adventure Activities
- Shopping

5. Budget & Expense Management

- Budget planning
- Expense recording
- Category-wise expenses
- Cost estimation
- Expense reports

Expense Categories

- Transportation
- Hotel
- Food
- Shopping
- Entertainment
- Miscellaneous

6. Group Collaboration System

- Create travel groups
- Invite members
- Shared itineraries
- Group discussions
- Shared expenses

- Role management

7. Destination Discovery System

- Browse destinations
- Destination details
- Attractions listing
- Travel guides
- Weather information
- Popular locations

8. Media & Document Management

- Upload travel documents
- Upload tickets
- Upload hotel bookings
- Upload travel photos
- Cloud storage

9. Notification System

- Trip reminders
- Activity reminders
- Budget alerts
- Group invitations
- Travel updates
- System notifications

10. Reports & Analytics

Traveler Dashboard

- Upcoming trips

- Budget overview
- Expense summary
- Travel statistics
- Favorite destinations

Admin Dashboard

- User analytics
- Trip analytics
- Destination analytics
- Revenue reports
- Platform statistics

11. Payment Management (Optional)

- Group expense splitting
- Shared payments
- Payment tracking
- Settlement management

12. Final Integration, Testing & Deployment

- Frontend and backend integration
- API validation and testing
- End-to-end workflow testing
- Security testing
- Performance optimization
- Docker containerization
- Production deployment
- Monitoring and logging setup
- Documentation and user guides

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Backend Setup

Tasks

- Define project scope and user roles.
- Design database schema:
 - Users
 - Roles
 - Trips
 - Itineraries
 - Activities
 - Destinations
 - Budgets
 - Expenses
 - Notifications
- Initialize Spring Boot project.
- Configure PostgreSQL / MySQL.
- Implement JWT authentication.
- Create React frontend skeleton.

Outcomes

- Backend and frontend architecture setup completed.
- Authentication flow functional.
- Database schema finalized.

- Role-based access configured.

Milestone 2: Week 3 & 4 — Trip & Itinerary Management

Tasks

- Implement trip management APIs.
- Build itinerary creation functionality.
- Develop destination pages.
- Create trip dashboards.
- Implement activity scheduling.

Outcomes

- Trip management operational.
- Itinerary workflow functional.
- Destination management completed.

Milestone 3: Week 5 & 6 — Budget, Expenses & Collaboration

Tasks

- Implement budget management system.
- Build expense tracking functionality.
- Create group collaboration workflow.
- Implement notifications.
- Add document upload system.

Outcomes

- Expense management operational.
- Group collaboration functional.
- Budget tracking completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build analytics dashboard.
- Add reports and charts.
- Implement testing and validations.
- Deploy frontend/backend using AWS, Render, or Vercel.
- Configure SSL and production environment.

Outcomes

- Fully deployed production-ready application.
 - Analytics and reporting systems operational.
 - Complete end-to-end travel planning workflow demonstrable.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Spring Boot project setup completed.
- JWT authentication implemented.
- Database schema finalized.
- Frontend authentication flow working.

Milestone 2 (Week 4)

- Trip management operational.
- Itinerary planning functional.
- Destination pages implemented.

Milestone 3 (Week 6)

- Budget management operational.

- Expense tracking functional.
- Collaboration workflow completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Analytics and reporting systems operational.
 - End-to-end travel planning workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend

- Java
- Spring Boot

Frontend

- JavaScript
- React.js

Database

Local Development

- MySQL

Production

- PostgreSQL

Authentication

- Spring Security
- JWT Authentication

- OAuth2 (Google Login)

Libraries & Frameworks

Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- Maven

Frontend

- React
- React Router
- Axios
- Tailwind CSS
- Context API
- Chart.js

External APIs

- Google Maps API
- OpenWeather API
- Stripe / Razorpay (Optional)

Notifications

- Firebase Cloud Messaging
- JavaMailSender

Testing

- JUnit

- Mockito
- Postman
- React Testing Library

Dev & Deployment Tools

- IntelliJ IDEA / VS Code
 - Git & GitHub
 - Docker
 - Docker Compose
 - GitHub Actions
 - AWS / Render / Railway
 - Postman
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8. Performance Metrics

Trip Management Metrics

- Trip creation success rate
- Itinerary completion rate
- Destination engagement rate
- Group collaboration participation rate

Budget & Expense Metrics

- Expense tracking accuracy
- Budget utilization efficiency
- Shared expense settlement accuracy

Travel Analytics Metrics

- Destination popularity tracking

- User engagement analytics
- Travel activity insights

System Performance Metrics

- API response time
 - Dashboard loading speed
 - Notification delivery success rate
 - Concurrent user handling capacity
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9. Example Quantitative Goals

Trip Planning

Enable users to create and manage detailed travel itineraries efficiently.

Budget Management

Provide accurate expense tracking and budget monitoring throughout trips.

Group Collaboration

Support collaborative travel planning and shared expense management.

Destination Discovery

Help users discover destinations, attractions, and travel insights effectively.

Platform Performance

Support multiple concurrent travelers and trip plans while maintaining stable performance and responsiveness.